

## 1. Overview

Our project centers around understanding the relationship between csRNA presence on the cell-surface and its expression in cell-cell interaction. Our procedure involves extracting csRNA from the cellular membranes of bone marrow-derived macrophages (BMDM), bone marrow-derived dendritic cells (BMDC), and activated M1-macrophages to isolate and map csRNA sequences of interest and filter them by relative abundance. Moving forward, we are interested in the classification of these sequences, the circumstances behind their expression, and how they are expressed in different cells. The long-term objective of this project is to identify which sequences play a significant role in interactions between cells, especially in inflammatory environments conducive to further immunology research.

## 2. Past Progress

Last semester was my first in the lab and participating on this project. I completed onboarding and assumed my role in developing a sequencing pipeline to extract and sort the most prevalent rRNA sequences from RNA samples found in an BMDM, BMDC, and M1 macrophage dataset. Standard BLAST and normalized pipelines are optimized for mRNA and DNA sequences, so I had to design my own. The pipeline we used normalized the M1 macrophage raw sequence counts to reads per million and then filtered out shared sequences on the BMDM and BMDC controls for M1 exclusivity. To identify the physical origins of these sequences, they are converted to FASTA and mapped using Bowtie2 against parent rRNA indexes. An issue within our procedure was that we could only extract sequences of limited read length; however, longer reads led to more significant results, so we performed contig assembly to turn <150nt reads into consensus sequences of 300-500nt. The end result of this process was a database of rRNA sequences sorted by relative abundance and filtered by exclusivity that can be mapped to specific regions on the cell surface.

## 3. Intent for Semester

Having identified exclusive sequences on a single M1 sample relative to two single controls, the objective now is to replicate the procedure on a much greater number of samples to determine if there is significance in our findings. My role is to apply my workflow to the expanded dataset and compare their results to identify trends and significance in our results, while optimizing the workflow. If we can confirm the presence of exclusive rRNA fragments on inflamed cells, then our interest turns to identifying the role these sequences serve and when or how cells will express them. One method to explore is locating these sequences' origins and designing probes to perform targeted experiments on their expression.

## 4. Continued Interest

A recent development in the project was the discovery of csRNA, with much more limited relative expression, on the cell membrane of red blood cells (RBCs). Calvin, another undergraduate student, and I will undertake the sequencing and decoding of these sequences on RBC samples. If csRNA sequences of interest are identified, then we will investigate the expression and relevance of such sequences and the role they may play in immunoregulation. Another avenue to investigate if RBCs contain significant csRNA is how their abundance or presence along the stages of erythropoiesis as RBCs mature.